10 Positron emission tomography (PET scanning) involves the detection of gamma-radiation in order to identify the position of origin of positrons in the body.

(a) (i) Positrons are not naturally present in the body.

Explain how positrons come to be present in the body during PET scanning.

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..... [2]

(ii) Explain how positrons cause the emission of gamma-radiation from the body during PET scanning.

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..... [3]

(b) Show that the wavelength of the gamma-radiation that is detected during PET scanning is approximately 2.4 pm. Explain your reasoning.

[4]

[Total: 9]